

ADVANCES IN HEART FAILURE, MECHANICAL CIRCULATORY SUPPORT AND TRANSPLANT

Node-Based Framework for Regionalized Cardiogenic Shock Systems of Care: Evidence, Design, and Future Directions

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ABSTRACT: Cardiogenic shock (CS) is a cardiovascular emergency with short-term mortality ranging from 30% to 40% despite an expanding therapeutic armamentarium. Structured systems of care have transformed outcomes in ST-segment–elevation myocardial infarction, stroke, and out-of-hospital cardiac arrest through tiered hospital designations, standardized protocols, and regionalized networks, yet no analogous framework exists for CS. CS differs from these conditions in ways that complicate direct application of existing models: the appropriate intervention varies by cause, hemodynamic phenotype, and shock severity; the therapeutic trajectory is iterative and uncertain rather than binary and time-fixed; prehospital identification is rarely feasible; and the exit strategy, encompassing myocardial recovery, durable mechanical support, transplantation, or palliation, has no parallel in other cardiovascular emergencies. Observational data suggest an association between structured multidisciplinary care, regionalized networks, and more consistent care delivery though causal relationships remain unestablished, and no standardized national framework for CS center designation yet exists. We propose that CS systems of care should be organized around a node-based model in which each hospital tier has a defined role, transfer is guided by physiological trajectory and exit strategy candidacy rather than geography, and system performance is evaluated by metrics of timing, outcomes, and equity. The American Heart Association Cardiogenic Shock Registry, designed to enroll consecutive patients across all etiologies and severity levels, represents the natural infrastructure around which this framework can be constructed. Realizing this vision will require dedicated registries, implementation research, and a definition of success that extends beyond in-hospital survival to encompass recovery, function, and equitable access to care.

Key Words: cardiopulmonary resuscitation ■ emergency cardiac care ■ shock, cardiogenic

Cardiogenic shock (CS) is a syndrome of tissue hypoperfusion resulting from inadequate cardiac output and represents a highly morbid and resource-intensive cardiovascular emergency.^{1,2} Despite an expanding therapeutic armamentarium that allows matching treatment with hemodynamic profile, outcomes have remained poor with short-term mortality ranging from 30% to 40% and 1-year mortality approaching or exceeding 50%.^{3–6} Beyond mortality, development of CS is associated with profound morbidity due to multiorgan dysfunction necessitating mechanical ventilation, renal

replacement therapy, acute stroke, and major bleeding or procedural-related vascular complications.

Central to persistently poor outcomes is the interplay between the rapid and dynamic nature of CS progression, timing of presentation, delays in recognition, and variability of specialized infrastructure for the management of critically ill and medically complex patients (Figure 1). Structured systems of care, encompassing early identification, standardized risk stratification, hemodynamic stabilization, and timely transfer to centers with advanced capabilities, have transformed outcomes in

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Nonstandard Abbreviations and Acronyms	
CS	cardiogenic shock
SCAI	Society for Cardiovascular Angiography and Intervention
STEMI	ST-segment–elevation myocardial infarction

other high-acuity cardiovascular emergencies, including ST-segment–elevation myocardial infarction, stroke, and out-of-hospital cardiac arrest. Whether analogous frameworks can be developed and implemented for CS, a condition defined by greater clinical complexity and heterogeneity, as well as a less predictable therapeutic trajectory, remains an important and evolving question. In this review, we examine the evidence base for existing cardiovascular systems of care and explore the unique

challenges CS poses to regionalization efforts. Building on prior expert consensus, we describe and provide justification for a node-based organizational framework that moves beyond unidirectional referral toward a dynamic network in which escalation and repatriation are governed by each patient's clinical trajectory, shock severity, and candidacy for advanced heart failure therapies.^{1,7}

ESTABLISHED SYSTEMS OF CARE FOR CARDIOVASCULAR EMERGENCIES

Structured systems of care have improved outcomes across several high-acuity cardiovascular conditions, establishing a framework of principles that may inform the development of analogous systems for CS. In ST-segment–elevation myocardial infarction (STEMI), the Mission: Lifeline initiative demonstrated that coordinating emergency medical services, referring hospitals, and

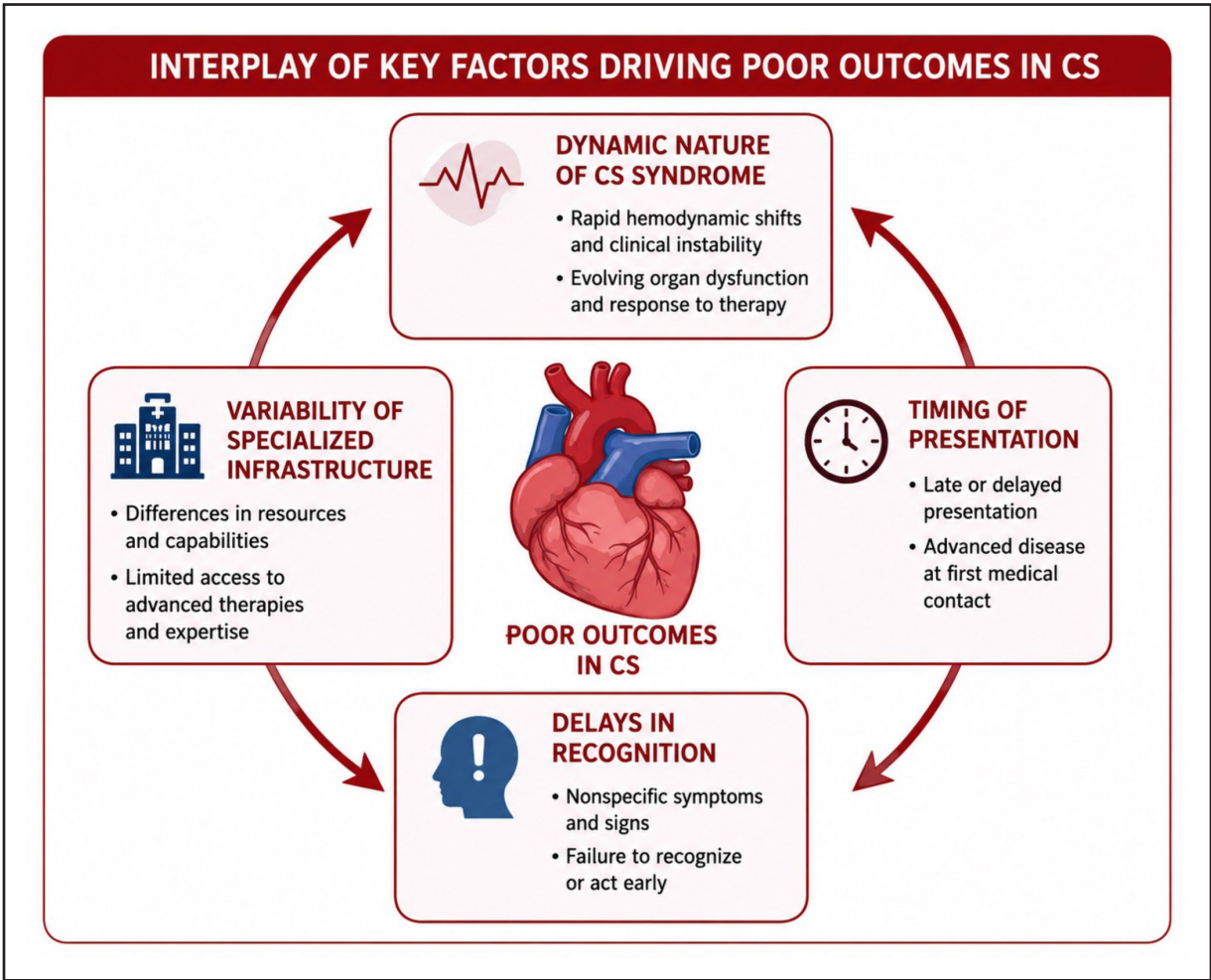


Figure 1. Drivers of persistently poor outcomes in cardiogenic shock (CS). Persistently poor outcomes in CS are driven by the interplay between the dynamic nature of the syndrome, timing of presentation, delays in recognition, and variability in specialized infrastructure required for the management of critically ill and medically complex patients. The dynamic and rapidly evolving hemodynamic profile of CS, coupled with delayed presentation and nonspecific clinical features, may impede timely diagnosis and initiation of advanced therapies. In addition, heterogeneity in institutional resources, expertise, and access to specialized multidisciplinary care contributes to variability in management strategies and clinical outcomes across healthcare systems.

percutaneous coronary intervention–capable centers into regional networks was associated with reduced door-to-device times and in-hospital mortality from 4.4% to 2.3% across 12 metropolitan areas, with >85% of the US population now covered by an STEMI system of care.^{8–10} Stroke systems have achieved comparable gains. The Target: Stroke program reduced median door-to-needle times from 78 to 50 minutes across 913 hospitals, with associated reductions in in-hospital mortality and improved rates of discharge to home, while a Canadian study of nearly 320 000 patients demonstrated a sustained 15% reduction in 30-day mortality in provinces with integrated stroke systems compared with those without.^{11,12} For out-of-hospital cardiac arrest, regionalized networks linking emergency services, bystander response programs, and specialized cardiac arrest centers have improved rates of neurologically intact survival, guided by the Chain of Survival framework that has served as an organizational template across diverse geographic settings.¹³

Across these conditions, effective systems share several structural features: tiered hospital designation based on resources and expertise, prehospital triage and notification protocols, standardized treatment algorithms, multidisciplinary team-based care, and systematic data feedback for quality improvement. Critically, however, the interventions driving benefit in STEMI, stroke, and cardiac arrest are time-sensitive, largely uniform, and binary in nature, namely, reperfusion, thrombolysis or thrombectomy, and defibrillation, respectively. CS presents a fundamentally different challenge: it is a dynamic and heterogeneous syndrome in which the appropriate intervention varies by cause, hemodynamic profile, severity of illness, and institutional capability; further, the therapeutic trajectory is far less predictable. In addition, evidence-based strategies for improving outcomes are not as readily available. Developing an effective CS system of care will, therefore, require adapting these established principles to a condition demanding not just rapid triage and transfer but also ongoing hemodynamic assessment, iterative treatment escalation, and access to a broad and evolving toolkit of advanced therapies.

EVIDENCE FOR CS SYSTEMS OF CARE

The evidence supporting structured systems of care specifically for CS has grown substantially though it remains predominantly observational and largely derived from single-center experiences, multicenter registries, and administrative databases. The foundational building block of CS systems of care is the multidisciplinary shock team. Single-center studies initially established proof of concept, demonstrating that a standardized, interdisciplinary team-based approach to CS was associated with lower in-hospital mortality.¹⁴ A subsequent multicenter analysis of cardiac intensive care units with

and without shock teams demonstrated that shock team centers were more likely to obtain invasive hemodynamic data, use advanced mechanical circulatory support beyond an intra-aortic balloon pump, and had lower risk-adjusted cardiac intensive care unit mortality, suggesting that the organizational infrastructure of shock care may influence outcomes independent of any single therapeutic decision.¹⁴ Importantly, a single-center registry of 1453 patients with consecutive CS evaluated the durability of this approach across 2 time periods (2017–2019 versus 2020–2022), finding that despite a shift toward higher acuity presentations, greater heart failure–related CS, and substantial changes in management including increased veno-arterial extracorporeal membrane oxygenation use and heart transplantation, standardized team-based care was associated with consistent in-hospital mortality over time, suggesting that systems' infrastructure may help maintain outcomes even as patient complexity evolves.¹⁵ Despite these advances, significant variability in care standards persists across institutions. A contemporary analysis of cardiac intensive care units demonstrated high variability in the use of invasive hemodynamic monitoring driven largely by institutional preference.¹⁶ However, the use of invasive hemodynamics to guide the management of CS was associated with lower mortality, a result that was corroborated in a separate contemporaneous registry and claims-based analyses from the National Inpatient Sample and Nationwide Readmissions Database.¹⁷

Beyond the shock team, the question of whether regionalization of CS care to higher-capability centers improves outcomes has gained increasing attention. Tehrani et al¹⁸ described a standardized, regionalized network of care across 34 centers and multiple hospital systems encompassing 3 states in the mid-Atlantic region and found that patients presenting initially to spoke hospitals had comparable short-term mortality and similar rates of bleeding, vascular, and stroke complications compared with those presenting directly to the hub, leveraging a shared protocol and structured transfer pathway. More recently, a large observational analysis of over 620 000 CS admissions from the Nationwide Readmissions Database demonstrated increased use of invasive hemodynamics and a graded association between hospital tier and survival: in-hospital mortality was lowest at referral centers or hubs and increased progressively with decreasing referral rates or spokes, from 29.5% at hubs centers to 45.2% at spoke centers, an association that persisted after risk adjustment and across clinically relevant subgroups.¹⁹ Critically, the survival advantage observed at hub centers differed between those that offered cardiac transplantation and durable left ventricular assist device services versus those that do not, but both of which possess advanced temporary mechanical circulatory support and

cardiothoracic surgical capabilities.²⁰ This may suggest that what distinguishes these programs is the broader institutional infrastructure of mature shock programs, including advanced heart failure expertise, early recognition of CS, multidisciplinary decision-making, and the capacity to navigate trajectories encompassing recovery, durable mechanical support, transplantation, and palliation. These observational findings collectively gave rise to the concept of tiered CS center designations, in which hospitals are categorized by available resources and expertise into a hierarchical framework designed to match patient acuity with institutional capability. The 2025 ACC Expert Consensus statement describes 3 levels of center capability ranging from initial diagnosis and stabilization (level 3) to comprehensive advanced heart failure and transplant programs (level 1) and provides foundational guidance on transfer. This review builds on that foundation to articulate a node-based organizational framework, one that moves beyond unidirectional referral toward a dynamic network in which escalation and repatriation are guided by clinical trajectory and candidacy for advanced heart failure therapies, such as transplantation and durable left ventricular assist device.¹

Despite this emerging signal, the evidence base for CS regionalization remains nascent, and several important gaps persist. Current frameworks for tiered CS center designation are based largely on expert consensus, and no standardized national classification system analogous to trauma systems of care yet exists.¹ Data pertaining specifically to the outcomes of transferred patients are limited and mixed: one multicenter registry analysis found higher mortality among transferred patients, driven primarily by heart failure–related CS, while other data suggest comparable outcomes when transfer occurs within a structured network.^{13,16} A recent single-center analysis of temporary mechanical circulatory support device–related outcomes showed substantially higher rates of bleeding, hemolysis, vascular injury, and mortality in patients implanted at a referral center before transfer to the hub.²¹ This disparity was attributed in part to presentations of more advanced CS at referral centers, underscoring the importance of early recognition and a focus on safe transfer for patients supported by devices. Importantly, existing observational data are largely time-blind: administrative data sets cannot capture when shock began, how rapidly patients deteriorated, what Society for Cardiovascular Angiography and Intervention (SCAI) stage they were at the time of transfer, or whether the transfer occurred before or after the window for meaningful intervention had narrowed or even closed. These limitations underscore that while the association between higher-tier care and improved outcomes is compelling, the mechanisms and the optimal design of CS systems of care remain to be defined prospectively.

DIFFERENT CONSIDERATIONS FOR CS

While CS shares surface-level features with STEMI, stroke, and out-of-hospital cardiac arrest, such as acuity, time-sensitivity, and the need for rapid escalation to definitive care, it differs from each of these conditions in ways that fundamentally complicate the design of a system of care. Understanding these distinctions is essential to avoid applying frameworks that were designed for conditions with a fundamentally different therapeutic logic.

The interventions that drive survival benefit in STEMI, stroke, and cardiac arrest are each defined by a single, largely binary therapeutic decision executed within a narrow time window: reperfusion, thrombolysis or thrombectomy, and defibrillation with targeted post-arrest care, respectively. In each case, the system's primary task is to deliver one treatment, to the appropriately selected patient, as rapidly as possible. CS does not conform to this model. The appropriate intervention varies by cause, hemodynamic phenotype, degree of ventricular involvement, and shock severity as classified by SCAI stage, and the decisions that matter most, namely, whether and when to escalate mechanical support, which device to use, when to transfer, and what the exit strategy should be, require iterative, multidisciplinary reassessments over hours to days rather than a single time-sensitive action.⁷ A further practical distinction is that, unlike STEMI or cardiac arrest, CS is often difficult to diagnose before hospital arrival. Most patients present to the nearest hospital, frequently a community hospital without advanced capabilities, and their true shock severity may not be apparent until end-organ dysfunction develops. Prehospital bypass to a higher-tier center is, therefore, rarely possible, and the system must instead be designed around recognition, stabilization, and timely transfer after initial presentation.

Perhaps, the most consequential distinction between CS and other cardiovascular emergencies is the concept of the exit strategy. In STEMI and cardiac arrest, the clinical question after the index intervention is predominantly focused on myocardial recovery: will myocardial and neurological function recover, and how can complications be minimized? In CS, the therapeutic trajectory is fundamentally uncertain at the time of presentation and may evolve over days to weeks. Temporary mechanical circulatory support cannot improve survival in the absence of either meaningful myocardial recovery or a definitive exit strategy. For patients in whom recovery is not anticipated and who are not candidates for heart replacement therapies, a well-defined transition to palliative care represents an equally important and distinct therapeutic destination. This means that the central clinical question in CS is not simply how fast can we intervene? but also what are we intervening toward? Answering the second question requires advanced heart failure expertise, multidisciplinary candidacy assessment, surgical input, and

longitudinal decision-making capabilities that extend well beyond the acute care environment and that are not uniformly available (Figure 2).

Finally, survivorship after CS represents a dimension of care that remains distinct and inadequately studied. Survivorship after CS likely shares features with patients experiencing post-intensive care unit syndrome, including physical, cognitive, and psychological impairment.^{22,23} However, survivorship after CS differs in clinical outcomes including mortality and rehospitalization. Among patients with CS who survive to hospital discharge, 1-year mortality approaches 50%, with excess mortality risk concentrated within the first 60 days post-discharge.²⁴ This differs substantially from general intensive care unit survivors, in whom mortality risk persists over years following discharge.²⁵ These features identify a unique experience associated with CS survivorship and warrant further investigation to improve clinical outcomes. Current clinical frameworks provide limited direction beyond the acute hospitalization, and postdischarge rehabilitation pathways are far less systematically integrated into

CS care than they are in post-STEMI or postcardiac arrest care. A mature CS system of care must, therefore, extend beyond the walls of the acute hospitalization and encompass structured longitudinal follow-up, goals-of-care planning, and coordination with advanced heart failure programs, a scope that existing cardiovascular emergency systems have not been designed to address.²⁶

A NODE-BASED MODEL FOR CS SYSTEMS OF CARE

The conventional hub-and-spoke model, adapted from STEMI and stroke systems of care, is organized around a single directional imperative: move the patient to the hub as quickly as possible so that a timely and definitive intervention may be delivered. This model is well-suited to conditions where the therapeutic answer is known, the intervention is uniform, and the hub is the only place capable of delivering it. Existing observational

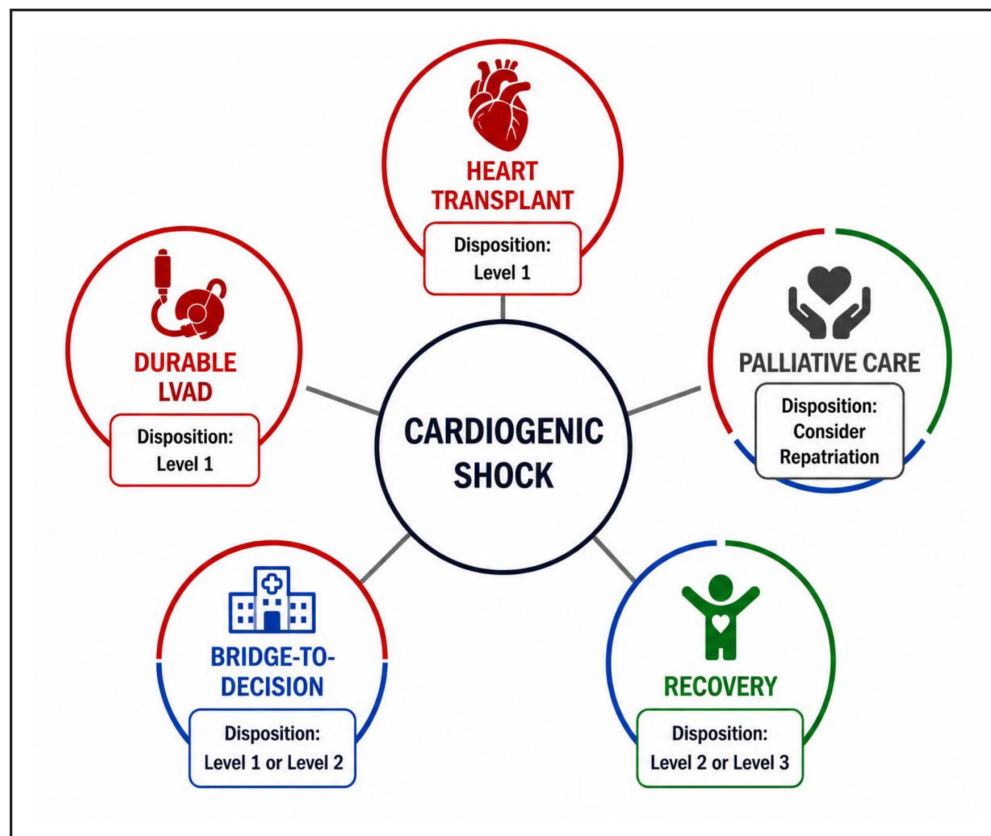


Figure 2. Exit strategy determines disposition.

Unlike other cardiovascular emergencies in which the therapeutic destination is often predefined, patients with cardiogenic shock may follow several distinct trajectories. The anticipated exit strategy helps determine the most appropriate disposition within a node-based system of care. Patients expected to recover may be managed within level 2 or level 3 centers, whereas those requiring durable left ventricular assist device (LVAD) therapy or heart transplantation generally require disposition to a level 1 comprehensive shock center. Patients undergoing bridge-to-decision evaluation may be managed within either level 1 or level 2 centers depending on clinical trajectory and local capabilities. Palliative care may be appropriate across all levels of the network, and repatriation to a center closer to home should be considered when consistent with patient goals and clinical circumstances. The figure illustrates how disposition decisions in cardiogenic shock are informed not only by current severity of illness but also by the anticipated trajectory and ultimate therapeutic destination.

data consistently associate higher-capability centers and multidisciplinary shock infrastructure with improved outcomes though the extent to which these findings reflect regionalization itself, patient selection, institutional expertise, or survivorship bias remains uncertain (Figure 3).

The clinical and organizational complexity of CS demands a systems model that goes beyond unidirectional transfer to a central hub and instead distributes defined roles, consultation capability, and bidirectional coordination across tiers. Indeed, not all patients with CS require or can safely reach a level 1 center, and reflexive

centralization risks overwhelming advanced programs with patients who could be appropriately managed at intermediate-tier hospitals. In addition, data from large administrative databases suggest that transfer alone, without a structured framework, does not reliably improve outcomes. Among patients with heart failure–related CS in the Nationwide Readmissions Database, transfer to hub centers was associated with higher in-hospital mortality compared with patients who were not transferred.¹⁹ Similarly, a Cardiogenic Shock Working Group analysis found higher mortality among patients transferred

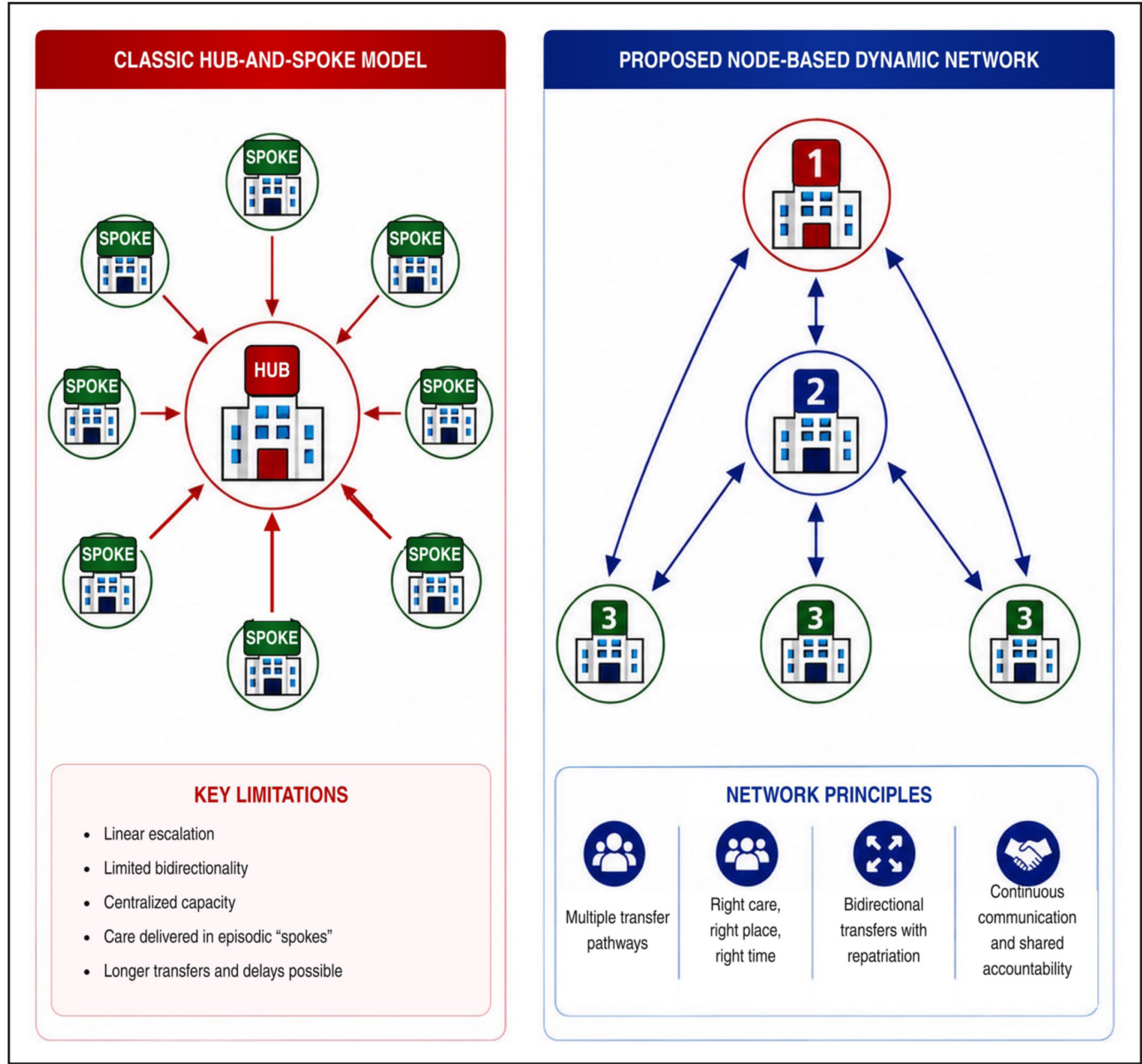


Figure 3. Node-based systems for cardiogenic shock (CS). Comparison of organizational frameworks for regionalized CS systems of care. The classic hub-and-spoke model (left) depicts a centralized referral structure in which patients are transferred from peripheral hospitals to a single high-capability center for advanced therapies, with limited bidirectional coordination. The proposed node-based dynamic network model (right) illustrates a flexible, interconnected system in which patient routing is determined by physiology, resource availability, and institutional capability rather than rigid hierarchical escalation pathways. Bidirectional coordination and recovery-oriented repatriation are emphasized to facilitate delivery of the right care at the right place and time across the continuum of CS management.

to tertiary care centers compared with those admitted directly to participating Cardiogenic Shock Working Group sites, which was driven primarily by patients with heart failure–related CS.²⁷ The longitudinal nature of CS care means that the hub cannot function as the sole locus of clinical decision-making, and a model that treats level 2 and level 3 hospitals purely as transfer conduits misses the essential role these institutions play in early recognition, initial stabilization, hemodynamic phenotyping, and the activation of transfer pathways before patients deteriorate beyond the point of rescue.

We propose that CS systems of care should instead be organized around a node-based model, in which each tier of the system has a defined, active, and accountable role rather than existing simply as a way station on the path to a higher-capability center (Figure 3). This concept has been articulated in the CS literature as an evolution beyond the hub-and-spoke framework, recognizing that effective regionalization requires distributed capability and bidirectional accountability rather than unidirectional transfer.^{7,28} In this framework, level 3 hospitals function as recognition and stabilization nodes: their core responsibilities are early identification of CS using standardized clinical and biochemical criteria, initiation of appropriate vasoactive support, bedside echocardiographic assessment where available, and prompt activation of and communication with a level 2 or level 1 center depending on clinical trajectory or likely exit strategy. Level 2 hospitals serve as advanced intervention nodes, capable of performing percutaneous coronary intervention when indicated, initiating invasive hemodynamic assessment, placing initial mechanical circulatory support, and managing patients in whom a local recovery pathway is plausible. Level 2 centers possess temporary mechanical circulatory support, associated surgical expertise, and maintain bidirectional relationships with level 1 centers for patients requiring cardiac replacement consideration, including complex advanced heart failure decision-making. Level 1 centers serve as regional coordination hubs, providing not only the highest level of clinical care but also real-time consultation, shared protocols, case review, education, and triage support for the broader network (Figure 4).

A critical feature of the node-based model is that transfer decisions are driven by physiology rather than geography. Escalation within a node system should be nonsequential. Patients are routed to the tier capable of delivering the care their anticipated trajectory requires, and transfer directly from a level 3 to a level 1 center is appropriate and expected when cardiac transplantation or durable left ventricular assist device evaluation may be needed. Movement through an intermediate tier is neither required nor desirable when that tier cannot provide adequate care that is required. The node model is, therefore, intended to reduce cumulative delay associated with hierarchical referral. Rather than automatic

transfer protocols based on hospital tier alone, escalation should be triggered by predefined clinical criteria reflecting deteriorating shock trajectory, including rising lactate, escalating vasoactive requirements, biventricular or right ventricular dominant shock, absence of a clear recovery pathway, or recurrent cardiac arrest. This physiologically guided escalation framework maps directly onto the SCAI shock stage classification, which provides a shared language for severity assessment across all tiers of the system and enables standardized communication between referring and receiving centers.²⁹ Importantly, the node model does not eliminate the need for transfer to level 1 centers; rather, it ensures that transfer occurs at the right time, for the right patient, before the window for meaningful intervention has closed.

The node-based model also demands a reconceptualization of what capability means within a tiered system. Capability should not be reduced to a checklist of available devices or procedural volumes, as administrative data sets require. A hospital that can implant a microaxial flow pump may not necessarily possess the hemodynamic expertise, device management protocols, vascular complication surveillance, organizational structure and staffing, and multidisciplinary decision-making infrastructure to optimize outcomes. The aspiration of a node-based system is, therefore, not to make every hospital a level 1 center but to ensure that every hospital understands its role, operates within a shared framework, and participates in the bidirectional accountability structures that make the system function as a network rather than a hierarchy.

Finally, a node-based model is better positioned than hub and spoke to address the equity dimensions of CS care. Geographic concentration of advanced CS capability in large urban academic centers creates access disparities that disproportionately affect rural, underinsured, and minority populations.^{7,28} The equitable delivery of CS care within a tiered system requires deliberate attention to factors that may be associated with differences in recognition, transfer, and outcomes. An analysis from the National Readmissions Database demonstrated that females had 27% higher odds of in-hospital mortality than males, despite being transferred at the same rates. This analysis suggests that structural inequities may exist within tiered systems, and that more studies are needed within tiered systems to reduce sex-based and other yet identified disparities³⁰ (Table).

CHALLENGES AND FUTURE DIRECTIONS FOR IMPLEMENTATION

A node-based system is not self-executing. Several barriers stand between the framework described above and its implementation, and each defines a corresponding research priority.

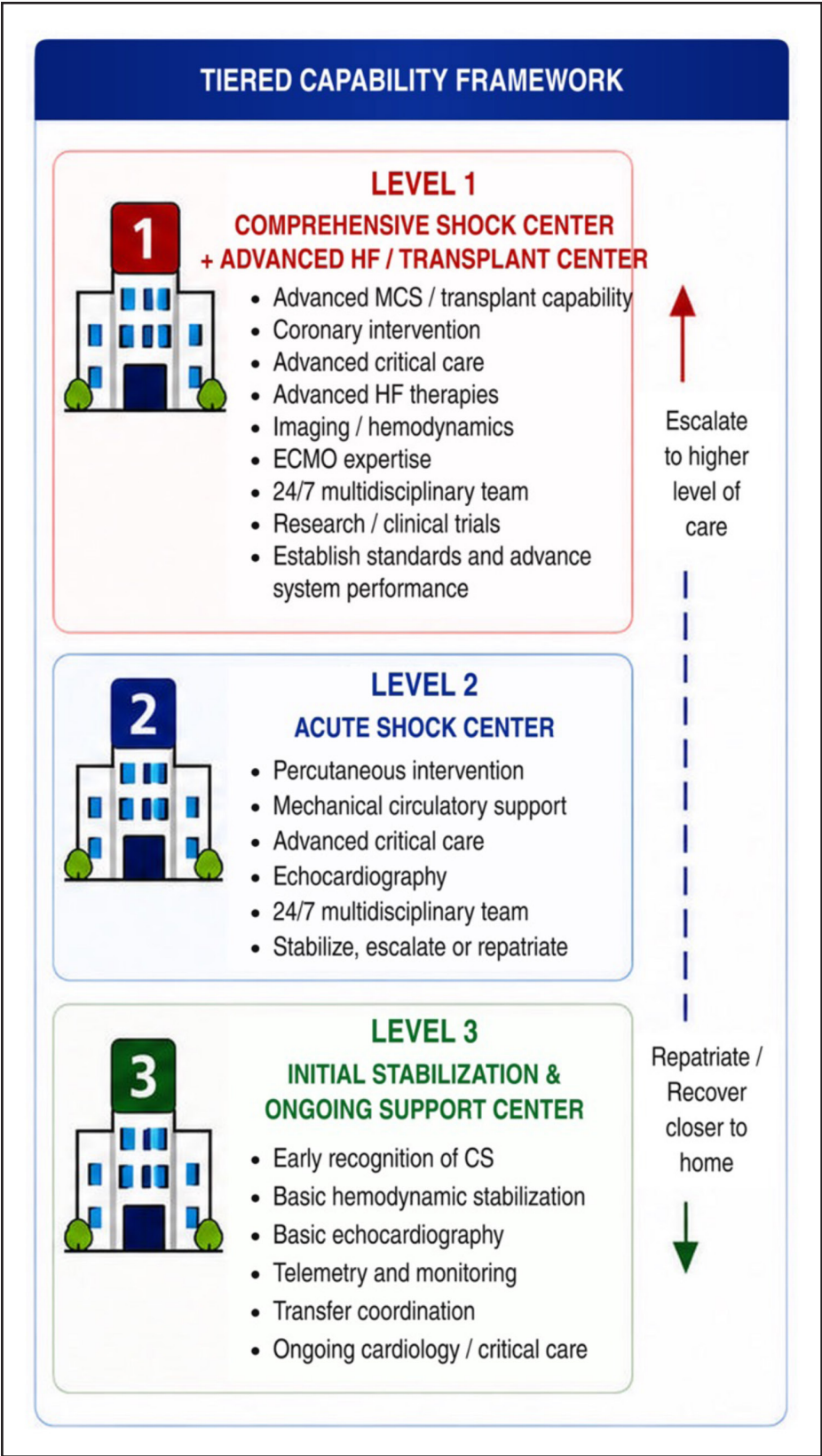


Figure 4. Tiered capability framework for cardiogenic shock (CS).

The refined tiered capability framework categorizes CS centers according to locally available expertise and infrastructure. Level 1 comprehensive shock centers provide advanced heart failure (HF) and transplant cardiology, durable left ventricular assist device therapy, cardiac transplantation, advanced temporary mechanical circulatory support, cardiac surgery, invasive hemodynamics, and dedicated shock team/cardiac intensive care unit expertise. Level 2 advanced shock centers provide invasive hemodynamic monitoring, (Continued)

The model depends on identifying, at or near presentation, which patients will require capabilities unavailable locally. Contemporary tools are poorly suited to this task. SCAI shock stage describes severity at assessment and is most prognostically informative once a patient has reached their most severe stage, but it was not designed to forecast which patients presenting at lower stages will progress.^{4,29} Shock stage alone is insufficient for risk assessment and must be interpreted alongside cause, hemodynamic phenotype, and nonmodifiable factors, including age and frailty, and whether staging captures trajectory rather than instantaneous severity remains unresolved.^{31,32} Patients who deteriorate after admission are those in whom triage decisions carry the greatest consequence, and they are least identifiable prospectively. Any system routing patients by anticipated need will, therefore, misroute some whose need is not yet apparent.

One mitigating principle is that consultative expertise should reach the patient before the patient reaches the center. Early contact with a level 1 shock team, through telemedicine, formal comanagement, or structured case review, allows trajectory to be assessed remotely and escalation to be triggered by deterioration rather than by completion of local evaluation. This is a partial solution. Developing dynamic risk models incorporating serial hemodynamic and biochemical data to predict trajectory rather than characterize current state is among the more consequential unmet needs in the field. Until such tools exist, tiered systems must be designed to tolerate prediction failure, with low thresholds for escalation, defined reassessment intervals, and system-level tracking of time from presentation to arrival at definitive care.

Exit strategy, defined as the intended destination of care following stabilization, presents a related difficulty. It should inform rather than rigidly dictate system architecture and must be continuously reassessed as the clinical picture evolves. Exit strategy categories should encompass myocardial recovery, bridge-to-decision, bridge to durable ventricular assist device or cardiac transplantation, and transition to palliative care for patients in whom further escalation is not consistent with clinical trajectory or patient values.³³

Candidacy is not established by hemodynamic criteria alone. It requires assessment of caregiver availability, psychosocial stability, adherence history, and financial circumstances through serial evaluations involving advanced heart failure clinicians, social work, and multidisciplinary committees and cannot be compressed into a single early conversation. Two features

of CS, however, make more of this work portable than is often assumed. Patients in SCAI stage D or E are frequently sedated or mechanically ventilated and cannot participate in psychosocial evaluation regardless of location, so assessment proceeds through collateral history obtained from caregivers. Those informants are rarely colocated with the level 1 center at presentation and are often more accessible near the referring hospital. Video-based caregiver interviews, remote social work consultation, virtual family meetings, and pretransfer verification of insurance status can, therefore, begin at the time of shock team activation rather than after arrival. This supports a distinction between provisional screening, which can identify absolute contraindications and recognize plausible candidates remotely, and definitive determination, which requires committee review, frailty assessment, and evaluation that does not translate to video. The binding constraint is, therefore, less often the patient location than the point at which the level 1 team is engaged. Embedding social work into initial shock consultation and defining a standardized psychosocial data set for collection at the referring center allows candidacy assessment to proceed in parallel with stabilization. Such approaches add unreimbursed workload, are complicated by interstate licensure requirements within networks spanning state lines, and have not been formally studied.

Repatriation presents a structural rather than clinical barrier. Return transfer preserves capacity at level 1 centers and allows recovery closer to family, yet the medical necessity criteria underwriting payer approval are asymmetrical, constructed to justify movement toward higher levels of care. A request to return a stabilized patient generally lacks the escalation rationale that approval requires. When coverage is denied, cost falls on the transferring institution or on families, or patients are retained at level 1 centers who no longer require that care. Palliative care, frequently relevant to this population, remains markedly underutilized, with <10% of CS admissions receiving formal services despite associations with lower readmission rates and reduced costs.^{34,35} Progress will require payer engagement to recognize repatriation as a reimbursable component of regionalized care and episode-based or network-level payment models in which financial consequences are borne across a system rather than by an individual hospital. Absent such change, repatriation will remain aspirational in many regions, and the capacity argument for a node-based system is correspondingly weakened.

Competing health systems within a single region introduce contractual incentives that discourage intersystem

Figure 4 Continued. percutaneous coronary intervention/revascularization, temporary mechanical circulatory support (MCS) capability, advanced cardiovascular critical care, and shock team expertise but do not offer cardiac transplantation. Level 3 initial diagnosis and stabilization centers provide early recognition, initial vasoactive therapy, basic hemodynamic stabilization, echocardiography/point-of-care ultrasound, and transfer coordination. ECMO indicates extracorporeal membrane oxygenation.

Table. Deficiencies of a Traditional Hub-and-Spoke Model for Cardiogenic Shock Systems of Care and Advantages of a Node-Based Framework

Limitation of traditional hub-and-spoke model	Why is this problematic in CS?	How a node-based model might address the limitation?
Assumes a single definitive intervention delivered at a central hub	Unlike STEMI or stroke, CS management is iterative and depends on cause, hemodynamic phenotype, shock severity, and response to therapy rather than a single time-fixed intervention.	Recognizes CS as a dynamic syndrome requiring longitudinal reassessment, multidisciplinary decision-making, and escalation pathways tailored to physiological trajectory
Organizes care around unidirectional transfer to a central center	Reflexive transfer may delay stabilization, overwhelm advanced centers, and fail to improve outcomes if transfer occurs too late in the shock trajectory.	Uses physiologically guided escalation and bidirectional coordination, ensuring transfer occurs for the right patient at the right time
Treats lower-tier hospitals primarily as transfer conduits	Community and intermediate hospitals play a critical role in early recognition, stabilization, invasive assessment, and activation of shock pathways before irreversible deterioration occurs.	Assigns active and accountable roles to every tier, including recognition, stabilization, intervention, escalation, and longitudinal coordination
Relies heavily on geography rather than physiology	Geographic routing alone does not account for shock severity, right ventricular failure, biventricular involvement, or candidacy for advanced therapies.	Transfer decisions are driven by predefined physiological triggers, including rising lactate, escalating vasoactive support, recurrent cardiac arrest, and lack of a recovery pathway.
Poorly suited for delayed or evolving recognition of shock	CS severity often evolves over hours, and many patients initially present to hospitals without advanced capabilities.	Emphasizes standardized early recognition, serial reassessment, and rapid consultation across the network
Does not adequately incorporate the concept of an exit strategy	Survival in CS depends not only on stabilization but also on recovery, durable LVAD, transplantation, or palliation.	Integrates advanced heart failure expertise, transplant evaluation, durable support planning, and goals-of-care discussions into regional care pathways
Defines capability primarily by device availability or procedural volume	Possession of temporary MCS devices alone does not ensure expertise in patient selection, anticoagulation, hemodynamic interpretation, or complication management.	Defines capability by multidisciplinary infrastructure, hemodynamic expertise, shock team maturity, and longitudinal care coordination
Limited bidirectional communication and shared accountability	Traditional referral systems may fragment care and reduce continuity between referring and receiving centers.	Establishes shared protocols, real-time consultation, education, case review, and collaborative network governance across all nodes
Focuses primarily on acute transfer and in-hospital survival	CS survivorship includes recurrent HF hospitalization, neurocognitive impairment, reduced quality of life, and ongoing advanced HF needs.	Extends the system beyond hospitalization to include longitudinal follow-up, rehabilitation, advanced HF care, and survivorship planning
Risks exacerbating disparities in access to advanced care	Concentration of advanced CS capability in urban academic centers may disadvantage rural, underinsured, and minority populations.	Promotes distributed expertise, telemedicine consultation, and equity-focused performance metrics including time-to-recognition and transfer access

CS, cardiogenic shock; HF, heart failure; LVAD, left ventricular assist device; MCS, mechanical circulatory support; and STEMI, ST-segment–elevation myocardial infarction.

transfer. Addressing this will require incentive structures rewarding appropriate transfer, shared governance in which competing systems participate in jointly administered regional networks, and regulatory engagement toward system-agnostic center designation.

Resolving these questions requires prospective evidence infrastructure capable of defining the functional responsibilities of each node and establishing tier-specific performance standards, which should reflect the distinct roles and resources expected of each level rather than being uniform across them. The American Heart Association Cardiogenic Shock Registry is the natural platform for this work.³⁶ By capturing time-stamped data on hemodynamic monitoring, mechanical circulatory support utilization, transfer timing, and outcomes across sites, it will empirically distinguish high-performing from lower-performing nodes, can serve as a neutral benchmarking platform that makes variation in transfer practice transparent across competing institutions, and could potentially support a learning health system model in which gaps are addressed through targeted quality improvement rather than blanket centralization. The American Heart Association Mission: Lifeline program demonstrates how registry infrastructure paired

with data feedback can transform care at scale through implementation science rather than any single therapeutic advance.^{8,9,37} Its longitudinal architecture should permit expansion of system performance beyond in-hospital survival to functional recovery, neurocognitive trajectory, and patient-reported quality of life, alongside equity metrics including time to recognition, transfer acceptance rates by hospital type and geography, and outcomes disaggregated by race, sex, and insurance status. None of these barriers argues against the framework. Each defines a testable implementation question, and together, they constitute an agenda for the prospective study of CS systems of care.

CONCLUSIONS

CS remains one of the most demanding clinical and organizational challenges in cardiovascular medicine. Observational evidence suggests that outcomes are influenced not only by available therapies but also by the systems within which care is delivered, with multidisciplinary shock teams and regionalized networks associated with more structured and consistent care delivery. A node-based model of regionalization, in which every tier

has a defined role, transfer is guided by physiological trajectory and exit strategy candidacy rather than geography, and system performance is evaluated by metrics of timing, outcomes, and equity, represents a rational and necessary evolution beyond the conventional hub-and-spoke framework. The American Heart Association Cardiogenic Shock Registry hopes to provide the infrastructure needed to validate this model, drive implementation science, enable tier-specific benchmarking, and expand the definition of system performance to encompass survivorship, equity, and patient-centered outcomes.

The next era of CS care will not be defined solely by newer devices but by whether healthcare systems can reliably match the right patient to the correct treatment at the appropriate time to maximize the likelihood of stabilization with an associated positive, patient-centered outcome. Realizing this potential will require prospective registries, shared governance across competing health systems, and a sustained commitment to implementation science modeled on the successes of STEMI and other cardiovascular emergency systems. The goal is not centralization for its own sake but the creation of adaptive, accountable networks capable of integrating acute rescue, exit strategy planning, survivorship care, and equitable access into a unified continuum of care.

ARTICLE INFORMATION

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